Error Decomposition in Training-Free Single-FLAIR Lower-Grade Glioma Segmentation: Generation versus Label-Free Selection

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Highlights

- Error decomposition separates localization, boundary representation, and selection.
- Expanded candidates raised oracle Dice from 0.872 to 0.906, not delivered Dice.
- The canonical pool covered 103/110 at Dice$\geq$ 0.70; 16/103 retained gaps $\geq$ 0.10.
- Weak tumour boundaries correlated with lower candidate ceilings across three cohorts.
- Frozen single-FLAIR fixed-slice segmentation achieved 0.712 Dice on external UCSF-PDGM.

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Abstract

Training-free single-FLAIR glioma segmentation can fail because the tumour is not localized, because its boundary is not represented by the generated candidates, or because an adequate candidate is available but not selected. We separated these error sources using a primary method with no learned weights that combines edge-preserving direction-dependent diffusion, score-topology threshold sweeping, and a closed-form label-free selection criterion; fuzzy α-cut candidates were added only in a retrospective capacity analysis. The whole-tumour target was evaluated on the ground-truth-selected largest-tumour axial slice from the TCGA-LGG development cohort (N=110), an independent UCSF-PDGM WHO grade 2–3 cohort (N=99), and BraTS 2023 (N=150) as a cross-grade stress test. Ground truth was used for slice selection, fixed-parameter development on TCGA-LGG, and retrospective evaluation, including candidate-pool ceilings and error decomposition; it did not enter case-level candidate generation or selection after the method was fixed. In TCGA-LGG, the label-free method achieved a mean Dice of 0.789, whereas the corresponding candidate-pool ceiling was 0.872. The tumour was localized in 106/110 cases, a candidate with Dice$\geq$ 0.70 was available in 103/110, and 16/103 representable cases retained a selection gap of at least 0.10. Expanding the candidate family under nested seed and α-cut budgets increased the ceiling to 0.906, with the final marginal gains satisfying a prespecified 0.005 practical-saturation margin; however, delivered Dice remained 0.789 under the existing label-free selector. The weak-boundary fraction was negatively associated with the pool ceiling in TCGA-LGG ($\rho = -0.537$), UCSF-PDGM ($\rho = -0.438$), and BraTS 2023 ($\rho = -0.467$), with the direction preserved after covariate adjustment. The frozen method achieved a mean Dice of 0.712 on UCSF-PDGM without retuning. These findings show that boundary-dependent candidate generation and ground-truth-free candidate selection are distinct bottlenecks. Reporting only the delivered mask obscures whether failure arises from localization, candidate-family capacity, or label-free decision-making.

Keywords: lower-grade glioma; FLAIR MRI; training-free segmentation; candidate generation; label-free selection; error decomposition; candidate-pool ceiling

1 Introduction

Glioma segmentation supports quantitative assessment of tumour burden and longitudinal change. Contemporary performance is dominated by supervised networks trained with expert masks and multiple co-registered MR sequences [1, 2]. Those conditions are not universal: retrospective archives may contain only one sequence, and expert delineations are costly. Fluid-attenuated inversion recovery (FLAIR) is particularly relevant in this restricted setting because it suppresses cerebrospinal fluid and highlights much of the whole-tumour abnormality. The scientific question addressed here is therefore not whether a handcrafted method can replace multi-modal supervised systems, but what can be achieved when only FLAIR is available and neither training nor learned weights are permitted.

Lower-grade glioma is a difficult instance of this problem. Its appearance varies substantially across patients, while infiltrative margins, partial-volume effects, and intensity inhomogeneity weaken both intensity and edge assumptions [3, 4]. A lesion may be almost isointense with surrounding parenchyma, and its boundary may be less conspicuous than unrelated anatomical interfaces. Consequently, a poor final mask does not identify a unique failure mechanism. The method may never reach the tumour; it may reach the tumour but fail to generate its boundary; or it may generate an adequate mask and then select a distractor. These failures require different remedies, yet a single delivered Dice score collapses them into one number.

We formulate this ambiguity as a three-stage error decomposition. *Localization* asks whether any candidate reaches a sufficient fraction of the tumour. *Boundary representation* asks whether the candidate family contains a mask with adequate overlap. *Selection* asks whether a closed-form, label-free criterion chooses such a mask when it is available. The best candidate identified retrospectively with ground truth is termed the *pool oracle*. It is a diagnostic measure of candidate-family capacity, not an inference output or a claim about the ceiling of all training-free methods. Joint reporting of the delivered mask and the pool oracle separates failure to generate from failure to select.

The primary method implements a generate–select strategy. A NewMetric direction-dependent diffusion front end produces an edge-preserving field from FLAIR [5]. Its delivered candidates are generated by seeded superlevel-set sweeping of a score field, and a label-free criterion ranks them using region quality and threshold persistence. Direct fuzzy α-cuts of intensity membership are introduced separately in the retrospective candidate-capacity analysis. Two intensity windows are evaluated within one common pool to accommodate bright-but-not-extreme lower-grade lesions without making tumour grade an inference-time input. NewMetric has a traceable Beltrami–Beil–Finsler derivation, but the subsequent traversal is performed on Euclidean image-grid superlevel components; the method is therefore not presented as Finsler-geodesic segmentation. Nor is improved accuracy uniquely attributed to the direction-dependent front end.

The evaluation deliberately separates inference from measurement. No labels or learned parameters enter candidate generation or selection. Ground truth is used to choose the largest-tumour axial slice and, retrospectively, to compute performance, pool ceilings, and error classes. The experiment consequently evaluates fixed-slice segmentation rather than automatic slice detection or 3D tumour localization. TCGA-LGG ($N = 110$) is the development and decomposition cohort; UCSF-PDGM WHO grade 2–3 ($N = 99$) provides frozen-parameter external validation; and BraTS 2023 ($N = 150$) is used only as a cross-grade stress test.

The contributions are fourfold. First, the study provides an operational localization–boundary–selection decomposition and reports it alongside label-free delivery. Second, it tests whether weak tumour boundaries are associated with candidate-family capacity across three cohorts. Third, nested seed and α-cut budgets quantify practical saturation of the candidate ceiling separately from selection accuracy. Fourth, matched candidate-channel, selector-robustness, and unsupervised-baseline controls define which gains belong to generation and which remain inaccessible to the label-free selector. The result is an evaluation framework for locating error in training-free candidate systems, rather than a claim of parity with supervised multi-modal segmentation.

2 Related Work and Study Positioning

Brain tumour segmentation studies occupy different supervision and input regimes, and their reported Dice values are not interchangeable. Supervised U-Net-type systems use pixel-wise labels and commonly combine T1, post-contrast T1, T2, and FLAIR [1, 2, 6]. Supervised single-FLAIR systems have also been evaluated on TCGA-LGG and related FLAIR data [7, 8], but their learned spatial features and labelled training protocol differ fundamentally from the present setting. At another point on the spectrum, SAM and MedSAM generate masks from box or point prompts [9, 10]. If such prompts are derived from the reference mask, the resulting score includes oracle localization; if prompts are generated automatically, prompt discovery becomes a separate localization task. These systems are therefore important performance references, but not matched comparators for prompt-free, training-free inference.

This distinction also motivates the term *training-free*. In this paper it means that inference uses neither a fitted predictive model nor learned weights; the handcrafted constants were nevertheless developed with reference-mask performance available on TCGA-LGG and then frozen. It is narrower than the often-used label-free or unsupervised designation, which may include representation learning or healthy-cohort anomaly models. Comparisons are consequently restricted by four protocol axes: supervision, modality, dimensionality, and target definition. The present protocol is training-free, single-FLAIR, two-dimensional, and targets the whole tumour on a ground-truth-selected slice.

Within the training-free regime, classical segmentation includes thresholding, clustering, seeded region growing, watershed methods, and active contours [3, 11]. These methods remain relevant because their decisions are traceable, but they depend strongly on intensity separability, initialization, and stopping rules. Multilevel Otsu- or entropy-based thresholding with metaheuristic search exemplifies single-mask intensity optimization [12]. Fuzzy clustering, originating from the fuzzy c-means formulation [13], and level-set combinations instead construct an initial region and evolve its boundary. Khosravanian et al. [14] reported high Dice on 40 selected BraTS 2017 FLAIR slices, whereas Darwish et al. [15] reported a high volumetric Dice on BraTS 2019. Those results were obtained under different sample-selection, dimensionality, and optimization protocols and are cited as method-family references rather than head-to-head performance estimates.

AUCseg is a particularly relevant unsupervised reference because its original three-dimensional, multiparametric framework first extracts whole tumour from T2-FLAIR by clustering before segmenting subregions with additional modalities [17]. The matched control used here reimplements only that whole-tumour concept under the present two-dimensional, single-FLAIR protocol; it is not a reproduction of the complete AUCseg system.

Candidate-pool methods expose an additional problem that single-mask methods hide. Biratu et al. [16] generated multiple automatically seeded region-growing candidates, but reported the best candidate selected against ground truth. This distinction is also consistent with recent work on ground-truth-free segmentation evaluation [18]. The best-candidate quantity is directly analogous to a pool oracle: it measures whether the generator can produce a suitable region, not whether a deployable label-free selector can identify it. The distinction is central here. We report both the label-free delivery and the retrospective pool oracle and retain zero-Dice cases in cohort means.

The primary and capacity-analysis candidate channels address complementary limitations. Seeded score-topology sweeping preserves connectivity to multiple plausible locations but can miss a lesion when no seed reaches it or when a low-contrast bridge causes leakage. Fuzzy α-cuts provide seed-independent, nested intensity-membership boundaries, but do not encode the plateau structure of the score field. Their union broadens representational capacity; it does not, by itself, solve candidate selection.

Beyond candidate construction, persistence supplies a principled language for stability across a filtration [19, 20]. In topological segmentation it may act as a shape prior through higher-dimensional holes or cavities [21]. Its role here is narrower: the connected component containing a seed is followed through score superlevel sets, and the width of a stable area plateau is recorded as a candidate attribute. This is a zero-dimensional, seed-conditioned stability measure, not a prior that specifies the expected topology of a glioma. Long persistence can indicate a well-separated region, but cannot establish that the region is tumour. Tumour specificity must therefore come from selection evidence and be tested empirically.

The literature commonly emphasizes candidate construction or final overlap, whereas fewer studies quantify whether a good candidate existed before it was selected. This omission matters because improving the generator can increase the number of plausible distractors and enlarge the selection problem. Conversely, a better selector cannot recover a boundary absent from the pool. The present study therefore treats pool capacity and label-free delivery as separate endpoints and uses nested candidate budgets to determine whether the observed ceiling reflects an undersampled family or a practical saturation of that family.

Taken together, the closest methodological comparators are not defined by their reported Dice alone, but by whether they are training-free, single-modality, prompt-free, and candidate-based. Within this space, the contribution is an error-accounting framework: localization failure, boundary-representation failure, and selection failure are assigned operational definitions and measured separately. The NewMetric front end provides a geometrically derived edge-preserving field, but the principal claim does not require Finsler-specific accuracy superiority. The novel empirical question is instead whether a candidate family can represent the tumour boundary and, conditional on that representation, whether an image-derived criterion can select it without ground truth. This positioning prevents pool oracles, selected-subset results, prompted outputs, and supervised models from being interpreted as equivalent forms of automatic training-free performance.

3 Materials and Methods

3.1 Study Design and Experimental Protocol

This study evaluated training-free segmentation of a prespecified axial FLAIR slice. The method received only the FLAIR image, required no learned parameters, and did not access the reference mask during candidate generation or label-free selection. Ground truth was used for five explicitly separated experimental purposes: selection of the largest-tumour slice, development of fixed parameters on TCGA-LGG, retrospective performance evaluation, diagnostic error decomposition, and computation of candidate-family ceilings. Consequently, the experiment addresses segmentation conditional on a tumour-bearing slice; it does not constitute end-to-end tumour detection or volumetric segmentation.

The cohorts had distinct and predefined roles. TCGA-LGG was the development and internal evaluation cohort. All fixed parameters were selected on this cohort with reference-mask performance available and were fixed before external evaluation. The WHO grade 2–3 subset of UCSF-PDGM was the independent lower-grade external-validation cohort. BraTS 2023 GLI was used as a cross-grade and preprocessing-domain stress test rather than as lower-grade validation. No parameter was re-estimated on either external cohort.

Two analyses must be distinguished. The *primary label-free method* denotes the fixed candidate-generation and selection procedure used to report delivered segmentation performance. The *expanded candidate-capacity analysis* enlarges the seed and fuzzy α-cut budgets in nested fashion and uses ground truth only to measure the best attainable candidate in each set. Its pool-oracle Dice is a retrospective upper bound on boundary availability, not a segmentation result available during label-free application. This distinction is maintained throughout the method, evaluation and results.

3.2 Imaging Cohorts and Reference Masks

TCGA-LGG [7] — *the focus set of the study*. Lower-grade (WHO II–III) gliomas, described using the contemporaneous cohort terminology [22], with low-to-moderate contrast and comparatively regular boundaries. The local analysis copy was derived from the public 256×256 three-channel TCGA-LGG derivative by extracting channel 1, the FLAIR channel. This extraction was verified against all source TIFF files; the pre-contrast and post-contrast channels were neither combined with nor supplied to the method. All tumour-bearing 2D axial slices were extracted (1360 *slices in total*); as required by the evaluation protocol, exactly **one** slice per patient — the one with the largest ground-truth tumour area — was retained, yielding an experimental set of 110 slices. These images are therefore described as a *skull-intact, preprocessed 2D derivative*, not as unprocessed DICOM. No additional skull-stripping or modality fusion was applied; a head-support mask was obtained by the threshold $I > 0.05$. All parameters were determined on this set alone. Table 1 summarizes the basic properties of the dataset.

Table 1: Descriptive statistics of the TCGA-LGG dataset (skull-intact preprocessed derivative, one largest-tumour slice per patient; 110 cases).

Property	Value
Original source	TCIA — TCGA-LGG collection [7]
Tumour type	Lower-grade glioma (WHO II–III)
Imaging sequence	FLAIR (single modality)
Preprocessing	Skull-intact preprocessed 2D derivative; study-specific scaling to [0, 1]
Number of patients	110
Tumour-bearing 2D slices per volume (all)	1 360 (min 3, max 36, mean 12.4)
Slice resolution	256×256 pixels
Head-mask ($I > 0.05$) area	~46–65% of the image area
Ground-truth (GT) tumour area — all 1360 *slices*	
Minimum	50 pixels
Maximum	7 929 pixels
Mean ± SD	1 950 ± 1 525 pixels
Median	1 541 pixels
Ground-truth (GT) tumour area — selected 110 *slices*	
Minimum	582 pixels
Maximum	7 929 pixels
Mean ± SD	3 013 ± 1 770 pixels
Median	2 642 pixels

BraTS 2023 (GLI) [23, 4, 24, 25] — *cross-grade stress test (no tuning)*. This cohort, of which only the FLAIR channel is used, consists predominantly of high-grade, skull-stripped and template-aligned cases. The same frozen settings were applied to the ground-truth-selected largest-tumour slices of 150 imaging studies from

143 patients. This cohort is not interpreted as lower-grade external validation; it probes operability under a different tumour grade and preprocessing regime.

UCSF-PDGM [26, 27] — *independent lower-grade external validation (no tuning).* UCSF-PDGM contains WHO grade 2–4 diffuse gliomas imaged with standardized 3T preoperative MRI together with expert-corrected multi-compartment tumour masks. Only the FLAIR channel of the WHO grade 2 (n=56) and grade 3 (n=43) cases was used here (N=99). Known follow-up/recurrence identifiers are grade 4 and do not enter this subset; the grade 2–3 identifiers are unique and none of them carries a BraTS21 identifier. For each case the axial slice with the largest ground-truth whole-tumour area was selected, and all parameters frozen on TCGA were applied unchanged. Because the UCSF images are skull-stripped, this cohort probes the cross-institutional lower-grade generalization of the primary candidate-generation and selection method; it does not independently validate the raw-image ROI correction.

Slice selection and the rationale for the 2D protocol. In all cohorts the unit is *a single axial slice per case*: among the slices of the volume, the one whose binary tumour mask has the largest pixel area is selected. This selection relies directly on the ground truth; the experiment is therefore not an end-to-end label-independent tumour detection protocol, and the reported values do not include automatic volume or slice finding performance. The ground truth is binarized in "whole-tumour" form by taking the union of all labels greater than zero.

There are three reasons for evaluating at the 2D slice level in this study: (i) the focus TCGA-LGG data are organized as per-patient selected 2D FLAIR slices; (ii) for protocol consistency, the same ground-truth largest-tumour-slice rule was applied to UCSF-PDGM and BraTS 2023; and (iii) 3D volumetric extension and automatic slice finding are outside the scope of this study. This protocol compares fixed-slice segmentation; it does not evaluate volumetric tumour detection.

3.3 Image Preparation and Score-Field Construction

Let I_{raw} denote the prespecified FLAIR slice. Intensities were scaled to the unit interval, a non-background support was defined, and values within that support were renormalized:

$$I = \frac{I_{\text{raw}}}{\max I_{\text{raw}} + \epsilon}, \qquad B = \{I > 0.05\}, \qquad I_n = \frac{I}{\max_B I + \epsilon}. \tag{1}$$

The NewMetric edge-preserving diffusion [5], which belongs to the broader family of edge-preserving anisotropic diffusion methods [28, 29], was applied with $\beta_{\text{NM}} = 5.0$, $\Delta t = 0.15$, and three iterations. Its result was rescaled within B to obtain

$$F_n = \frac{\text{NewMetric}(I) - \min_B \text{NewMetric}(I)}{\max_B \text{NewMetric}(I) - \min_B \text{NewMetric}(I) + \epsilon}. \tag{2}$$

Gaussian filtering was not used in the segmentation signal path.

The Sobel gradient magnitude $G = \|\nabla F_n\|$ was converted to an edge permeability field using the intra-support 95th percentile $K = P_{95}(G; B)$:

$$g = \frac{1}{1 + (G/K)^2}. \tag{3}$$

For each of two complementary intensity windows, a soft band-pass weight and a score field were defined as

$$w(I_n; \tau_h) = \frac{1}{1 + \exp[-(I_n - 0.28)/0.05]} \frac{1}{1 + \exp[(I_n - \tau_h)/0.05]}, \tag{4}$$

$$S = g F_n \mathbf{1}_B w, \tag{5}$$

where $(\tau_h, p_h) \in \{(0.82, 88), (0.97, 98)\}$. Here p_h is the upper FLAIR percentile used for seed eligibility. To avoid favouring the window from which a candidate originated, final candidate assessment used the window-independent field $E = g F_n \mathbf{1}_B$.

3.4 Primary Candidate Generation and Label-Free Selection

Seed localization and region evolution were deliberately based on different evidence. Seed locations were local maxima of S restricted to $p_{30}(I_n) \le I_n \le p_h(I_n)$. At most ten seeds were retained per window, with a minimum separation of 12 pixels. For each seed s, the threshold was decreased in 200 equally spaced levels from $S(s)$ to $0.20S(s)$, and the connected component containing that seed was followed:

$$C_t(s) = \mathrm{CC}_s\left(\{S \geq t\} \cap B\right). \tag{6}$$

Evolution was terminated when the component exceeded 25% of B, exhibited a fourfold area jump, satisfied the predefined acceleration condition, or met the edge-wall condition. Successive components whose relative area change did not exceed 0.06 formed one stable interval. Candidate recording began after the component reached $\max(40, 0.02|B|)$ pixels; the duration of its stable interval defined persistence $\pi(M)$. Thus, seeds were determined from a FLAIR-restricted score maximum, whereas candidate boundaries were determined by the topology of score superlevel sets.

For every candidate M, label-free quality combined area, mean evidence, compactness and solidity:

$$Q(M) = |M|\,\overline{E}_M\,\frac{4\pi|M|}{P(M)^2}\,\frac{|M|}{|\mathrm{hull}(M)|}. \tag{7}$$

Candidates from both intensity windows were considered jointly, and the delivered segmentation was

$$\widehat{M} = \arg\max_{M \in C_{\text{primary}}} Q(M)\pi(M)^{\gamma}, \qquad \gamma = 1. \tag{8}$$

Persistence rewards components that remain stable over a wider threshold interval; it does not, by itself, establish tumour specificity. The exponent was selected on TCGA-LGG and fixed before external evaluation.

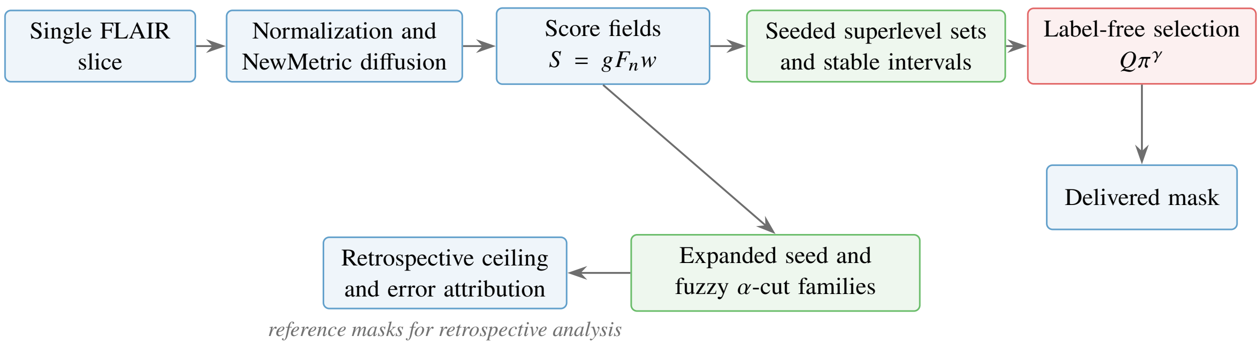

Figure 1: **Study method and analytical separation.** The upper sequence produces the label-free segmentation. Expanded seed and fuzzy α-cut families are evaluated only to measure candidate capacity and attribute errors; they do not replace the primary delivered mask. Reference masks were also used earlier for TCGA parameter development and prespecified slice selection, as detailed in Section 3.1.

3.5 Conditional Brain-ROI Renormalization

In skull-intact TCGA images, bright extracranial structures can determine the normalization maximum and compress intra-brain tumour contrast. A morphological brain ROI was therefore estimated from the image, with peripheral bright tissue removed and orbital regions excluded when their dark-centred, approximately circular and lateral configuration satisfied the predefined anatomical rules. The score fields and primary candidate family were then reconstructed within this ROI. Let M_0 and M_1 denote the masks selected before and after ROI renormalization, respectively, and let $f_{\text{in}}(M) = |M \cap R|/|M|$. The ROI-renormalized mask was adopted only if

$$f_{\text{in}}(M_0) < 0.65 \quad \text{and} \quad f_{\text{in}}(M_1) \geq 0.50; \tag{9}$$

otherwise M_0 was retained. The condition is inactive for already skull-stripped images and was therefore not independently validated by the two external cohorts.

The two adoption thresholds were developmental choices. Their sensitivity was audited by varying the first-mask threshold over 0.55–0.75 and the ROI-candidate threshold over 0.40–0.60. Tumour size and peripheral location were examined retrospectively using the upper quartiles of reference-mask area and normalized reference-centroid eccentricity; these quantities were not inputs to the gate.

3.6 Expanded Candidate-Capacity Analysis

The expanded analysis tested whether a suitable boundary was absent from the candidate family or present but not recoverable by label-free ranking. In addition to seeded superlevel-set candidates, four-centre one-dimensional fuzzy c-means [13] was applied to intra-support intensities. The darkest class was excluded, and connected components of fixed membership α-cuts were added as seed-independent boundary hypotheses. These candidates were used for capacity measurement, not for the primary delivered segmentation.

Nested seed budgets $s \in \{10, 15, 20, 25, 30\}$ and nested α-level counts $a \in \{0, 6, 11, 15, 23, 39\}$ were evaluated without removing candidates admitted at smaller budgets. For case i,

$$U_i(s,a) = \max_{M \in C_{i,s,a}} D(M, G_i), \qquad U(s,a) = \frac{1}{N} \sum_{i=1}^{N} U_i(s,a), \tag{10}$$

where G_i is the reference mask. U is therefore a retrospective candidate-family ceiling. Practical saturation required the upper bounds of the paired bootstrap 95% confidence intervals for both final budget increments to be smaller than the prespecified margin $\varepsilon = 0.005$ Dice.

Computational cost was measured over all 110 TCGA slices on an Intel Core Ultra 7 265K CPU (20 physical cores, 31.6 GiB RAM, Windows 11). The full canonical timing included NewMetric preprocessing, candidate generation and label-free selection. The expanded-budget timing was recorded separately and measures preprocessing plus the additional candidates through 30 seeds per window and 39 nested α levels; it loads the frozen canonical pool and is therefore not a second end-to-end inference time. The worker count used by the historical canonical batch run was not stored in its provenance, whereas the expanded-budget audit was run serially; the reported values are therefore descriptive computational costs rather than standardized serial-latency benchmarks.

3.7 Error Decomposition and Evaluation Measures

For each case, candidate localization was summarized by $R_{\max} = \max_j |M_j \cap G|/|G|$, and boundary availability by the pool-oracle Dice $D_{\max} = \max_j D(M_j, G)$. Errors were assigned hierarchically: $R_{\max} < 0.50$ indicated localization failure; $R_{\max} \geq 0.50$ and $D_{\max} < 0.70$ indicated boundary-representation failure; $D_{\max} \geq 0.70$ with $D_{\max} - D(\widehat{M}, G) \geq 0.10$ indicated selection failure; all remaining cases were classified as near-ceiling/successful. These thresholds are operational diagnostic definitions, not clinical decision limits. Descriptive sensitivity was assessed over the complete $3 \times 3 \times 3$ grid $R_{\max} \in \{0.40, 0.50, 0.60\}$, $D_{\max} \in \{0.60, 0.70, 0.80\}$, and selection-gap threshold $\in \{0.05, 0.10, 0.15\}$. This audit changed only the retrospective class labels, not the predicted masks.

The primary performance measure was Dice, calculated over all cases including complete misses. Jaccard overlap, sensitivity, precision, the number of zero-Dice cases, and the numbers exceeding Dice thresholds of 0.50 and 0.70 were secondary measures. Boundary agreement was characterized by average symmetric surface distance, 95th-percentile Hausdorff distance, and boundary F1 within two pixels; distance measures were summarized only when the two masks overlapped [30]. Lesion-level sensitivity, precision and F1 used connected-component matching at IoU ≥ 0.10.

Boundary strength was measured in inner and outer bands of the reference contour. The primary descriptor was the fraction of boundary locations whose NewMetric gradient magnitude fell below the intra-support median. Additional descriptors included normal-direction gradient magnitude and sign consistency, inner–outer ring intensity separation, tumour area and compactness. These quantities used ground truth only for retrospective error attribution.

3.8 Statistical Analysis

The observational unit was one prespecified axial slice per imaging study. TCGA-LGG and UCSF-PDGM contributed one study per patient. The 150 BraTS studies represented 143 unique patients; repeated studies were kept in the same resampling cluster. Means included zero-Dice cases. Percentile bootstrap 95% confidence intervals used 20,000 resamples. Case-level differences were resampled for paired comparisons, whereas boundary analyses used patient-clustered resampling. The corresponding fixed random seeds were 20260715 for primary paired analyses and 20260723 for cross-cohort boundary analyses.

Spearman correlations quantified the association between boundary descriptors and candidate ceilings; partial Spearman analyses controlling for area, compactness and ring contrast used 5,000 clustered resamples. The UCSF grade comparison was exploratory and used a two-sided Mann–Whitney U test, Cliff's δ [31] and an independent bootstrap interval for the mean difference. Parameter sweeps and all TCGA-based parameter comparisons were treated as developmental rather than confirmatory.

Table 2: Fixed parameters of the primary label-free method and the retrospective capacity analysis.

Stage	Parameter	Fixed value
Normalization	Non-background support	$I > 0.05$
NewMetric	β_{NM}, Δt, iterations	5.0, 0.15, 3
Edge field	Gradient and scale	Sobel magnitude; intra-support P_{95}
Intensity weighting	Lower bound; softness	0.28; 0.05
Intensity weighting	Upper windows	0.82 and 0.97
Seed localization	FLAIR percentile bands	$[30, 88]$ and $[30, 98]$
Seed localization	Maximum per window; spacing	10; 12 pixels
Region evolution	Threshold range; levels	$S(s)$–$0.20S(s)$; 200
Region evolution	Area cap; jump ratio	$0.25\|B\|$; 4
Stable interval	Relative area tolerance	0.06
Candidate eligibility	Minimum area	$\max(40, 0.02\|B\|)$ pixels
Selection	Persistence exponent γ	1
ROI condition	Adoption thresholds	$f_{\mathrm{in}}(M_0) < 0.65$, $f_{\mathrm{in}}(M_1) \geq 0.50$
Capacity analysis	Seed budgets	10, 15, 20, 25, 30 per window
Capacity analysis	Nested α counts	0, 6, 11, 15, 23, 39
Saturation	Practical Dice margin	0.005

4 Results and Discussion

4.1 Primary Segmentation Performance

On the TCGA-LGG development cohort, the primary label-free method achieved a mean Dice of 0.7893 and a median of 0.9134 over all 110 cases; six cases had zero Dice. Dice was at least 0.70 in 88/110 cases. These values describe the two-window, ten-seed-per-window superlevel-set candidate family selected by $Q\pi^{\gamma}$ with $\gamma = 1$; they do not include fuzzy α-cut candidates or ground-truth oracle selection.

The same primary candidate family contained 53,002 masks in total, with a median of 489.5 candidates per case. Retrospective oracle selection raised the mean Dice to 0.8722, leaving a mean oracle–selected difference of 0.0829 (Table 3). The oracle value is an attainable ceiling within the generated family and is not a label-free segmentation result. The coexistence of a high median delivered Dice and a non-negligible mean oracle gap indicates a heterogeneous error distribution: most cases were segmented well, whereas a smaller subset incurred either candidate-generation or candidate-selection failure.

4.2 Localization–Boundary–Selection Error Decomposition

The hierarchical analysis localized the source of error more precisely than the cohort mean (Table 4). Candidate localization succeeded in 106/110 cases, and a candidate with Dice ≥ 0.70 was available in 103/110. Thus, only three of the 106 localized cases were classified as boundary-representation failures. Among the 103 boundary-representable cases, 16 retained an oracle–selected difference of at least 0.10 and were therefore classified as selection failures. The remaining 87 cases were near-ceiling/successful.

The 27-setting threshold audit yielded 3–5 localization failures, 2–11 boundary-representation failures, 12–19 selection failures, and 81–87 near-ceiling cases. Canonical counts were unchanged when the oracle threshold

Table 3: Primary TCGA-LGG result and retrospective capacity of the same candidate family. All means include zero-Dice cases.

Measure	Result
Label-free mean / median Dice	0.7893 / 0.9134
Zero-Dice cases	6/110
Cases with Dice ≥ 0.70	88/110
Median candidates per case	489.5
Pool-oracle mean Dice	0.8722
Mean oracle–selected difference	0.0829

was varied from 0.60 to 0.70 and when the selection-gap threshold was varied from 0.10 to 0.15; the largest redistribution occurred at the stricter oracle threshold of 0.80. Selection failures outnumbered localization and boundary failures throughout the grid, although the exact counts remained definition-dependent.

Table 4: Hierarchical error decomposition on TCGA-LGG. Reference masks were used only retrospectively for attribution.

Error class	Cases	Selected Dice	Oracle Dice
Localization failure	4	0.084	0.115
Boundary-representation failure	3	0.166	0.437
Selection failure	16	0.394	0.844
Near-ceiling / successful	87	0.916	0.927

These results identify two distinct residual limitations. Candidate generation was insufficient in seven cases: four were not adequately localized and three were localized but lacked a boundary candidate meeting the representability criterion. In a larger group, however, an adequate candidate already existed and was not selected. Accordingly, increasing candidate diversity and improving label-free ranking address different scientific problems; a higher candidate ceiling cannot be interpreted as improved delivered segmentation unless the selection rule can recover that additional capacity.

Figure 2 illustrates the operational meaning of the four classes using representative cases chosen near the median selected Dice within each class. Candidate reachability may fail before a plausible tumour boundary is generated; alternatively, an adequate pool-oracle boundary may be present but rejected by the label-free selector. The near-ceiling example shows agreement between the available oracle boundary and the delivered mask.

To balance the failure-oriented audit with the method's typical successful behaviour, Figure 3 shows four cases from the near-ceiling/successful class. The cases were selected reproducibly by dividing that class into quartiles of reference-tumour area and, within each quartile, choosing the case closest to the class median selected Dice (0.932). This rule avoids selecting visually favourable extremes while spanning a fourfold range of lesion area. Only the genuinely label-free selected boundary is shown against the reference; no pool-oracle mask enters the displayed method output.

4.3 Candidate-Channel Contribution and Practical Saturation

The matched channel analysis confirmed this separation (Table 5). The seeded superlevel-set family alone delivered 0.7893 Dice and had an oracle ceiling of 0.8722. The fuzzy α-cut family alone was weaker in delivered performance (0.6473) and oracle capacity (0.8010). Adding it to the seeded family increased the oracle ceiling by 0.0275, from 0.8722 to 0.8997, eliminated cases with zero oracle overlap, and raised Dice-≥ 0.70 candidate coverage from 103/110 to 107/110. In contrast, label-free delivered Dice changed by 0.0000 and remained 0.7893. Thus, the α-cut channel produced a demonstrable gain in candidate availability, but the existing selector did not recover that added capacity as improved delivered segmentation.

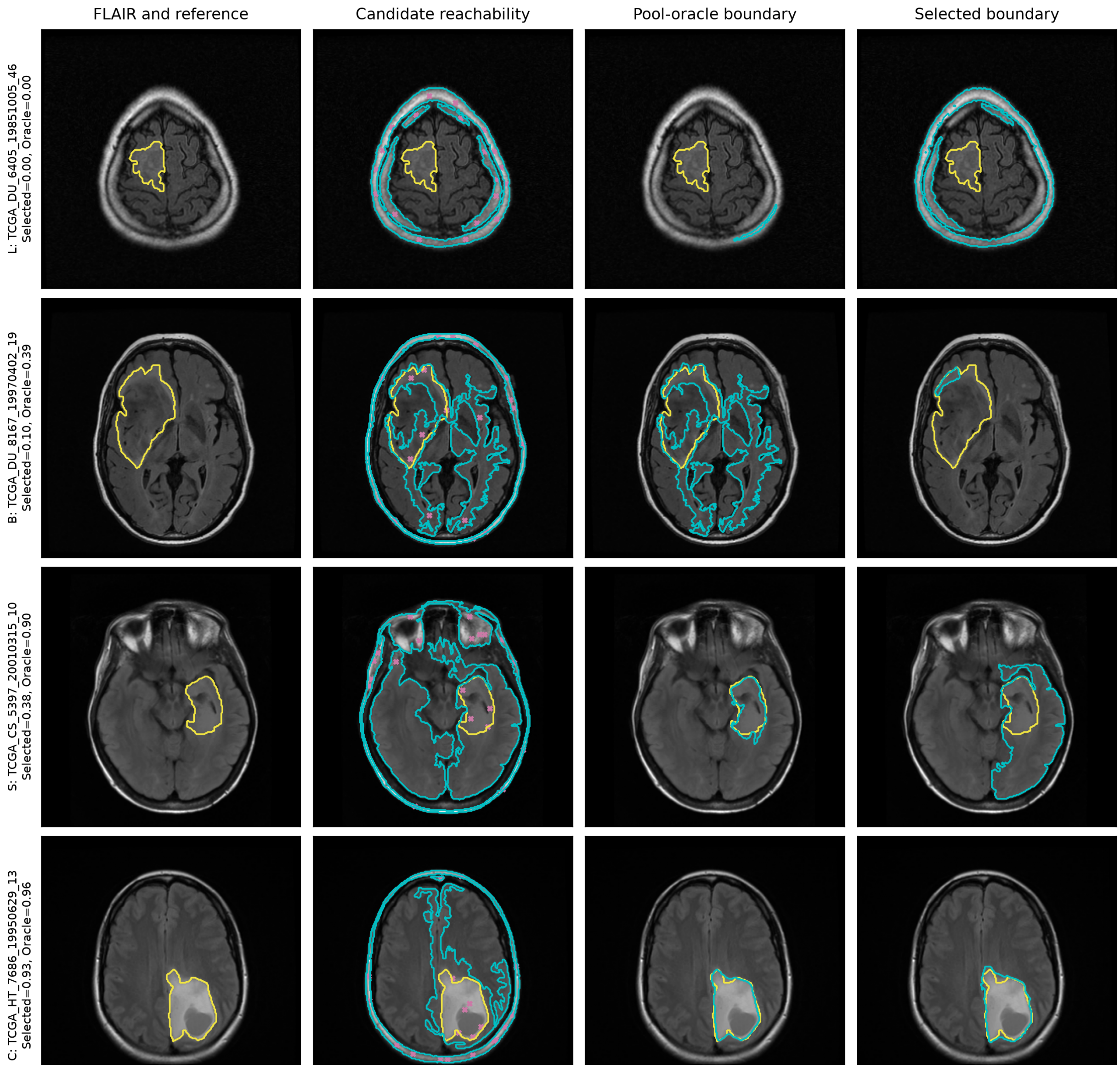

Figure 2: **Representative cases from the localization–boundary–selection decomposition.** This is a retrospective diagnostic audit, not the inference workflow: the pool-oracle column uses the reference mask only to identify the best available candidate. Rows show localization failure (L), boundary-representation failure (B), selection failure (S), and a near-ceiling/successful case (C). Columns show the FLAIR slice with the reference contour, the union of generated candidates and seed locations, the retrospective pool-oracle boundary, and the boundary selected without reference labels. Yellow denotes the reference mask, cyan denotes the indicated candidate boundary, and magenta crosses denote seed locations. Reference information and oracle selection are displayed only for retrospective interpretation.

Table 5: Matched candidate-channel analysis on TCGA-LGG. Oracle values require the reference mask and are not label-free performance.

Candidate family	Median count	Delivered Dice	Oracle Dice	Oracle zero
Seeded superlevel sets	489.5	0.7893	0.8722	3
Fuzzy α-cuts	159.0	0.6473	0.8010	0
Union	648.5	0.7893 (+0.0000)	0.8997 (+0.0275)	0

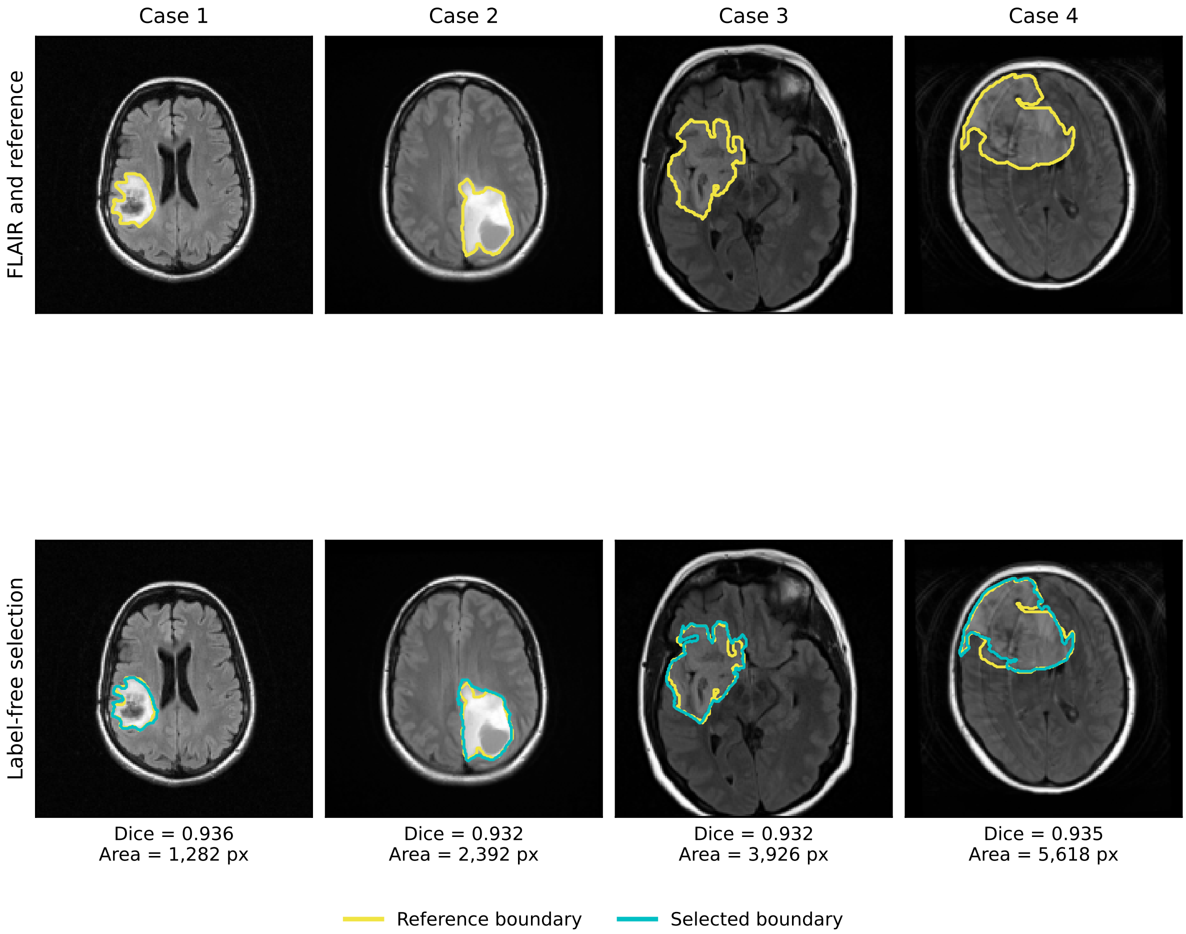

Figure 3: **Representative successful label-free segmentations across lesion-size strata.** Columns show one TCGA-LGG case from each quartile of reference-tumour area within the near-ceiling/successful class. The top row shows FLAIR with the reference boundary; the bottom row overlays the boundary selected without reference labels. Cases 1–4 are TCGA_FG_7643_20021104_26, TCGA_HT_7686_19950629_13, TCGA_FG_8189_20030516_25, and TCGA_CS_5396_20010302_15, respectively. Yellow denotes the retrospective reference and cyan the label-free selected boundary.

Increasing the nested seed and α budgets raised the oracle ceiling from 0.8722 to 0.9055, a paired mean increase of 0.0333 (95% CI $[0.0080, 0.0661]$), and provided a Dice-$\geq$ 0.70 candidate in $108/110$ cases (Table 6). The mean gain from the final seed increment ($25 \to 30$) was 0.000025 with an upper 95% confidence bound of 0.000075; the final α increment ($23 \to 39$ levels) produced no further mean gain. Both upper bounds were below the prespecified 0.005 Dice margin. The examined family therefore reached practical saturation near 0.906. This finite experiment does not establish a universal single-FLAIR limit or preclude a different candidate generator from attaining a higher ceiling.

The full canonical method required a mean of 1.659 s per slice (median 1.688 s; IQR 1.508–1.821 s). The separately timed expanded-budget workload required a mean of 3.984 s per slice (median 4.059 s; IQR 3.339–4.474 s). The latter is an additional research-analysis cost with a different scope and should not be added to or compared directly with the end-to-end canonical time. Because concurrency was not preserved in the historical canonical provenance, neither value is presented as a deployment latency benchmark.

4.4 Boundary Strength and Attainable Accuracy

The fraction of weak locations along the reference tumour boundary was negatively associated with the candidate-pool ceiling in all three cohorts (Table 7). Spearman correlations were −0.537 in TCGA-LGG, −0.438 in UCSF-PDGM grade 2–3, and −0.467 in BraTS 2023. After adjustment for tumour area, compactness and ring

Table 6: Nested candidate-capacity analysis on TCGA-LGG.

Candidate budget	Oracle Dice	Difference	95% CI
10 seeds, no α-cuts	0.8722	—	—
15 seeds, no α-cuts	0.9010	+0.0288	—
10 seeds, six α levels	0.8997	+0.0275	—
15 seeds, six α levels	0.9048	+0.0326	—
30 seeds, 39 α levels	**0.9055**	**+0.0333**	$[\mathbf{0.0080}, \mathbf{0.0661}]$

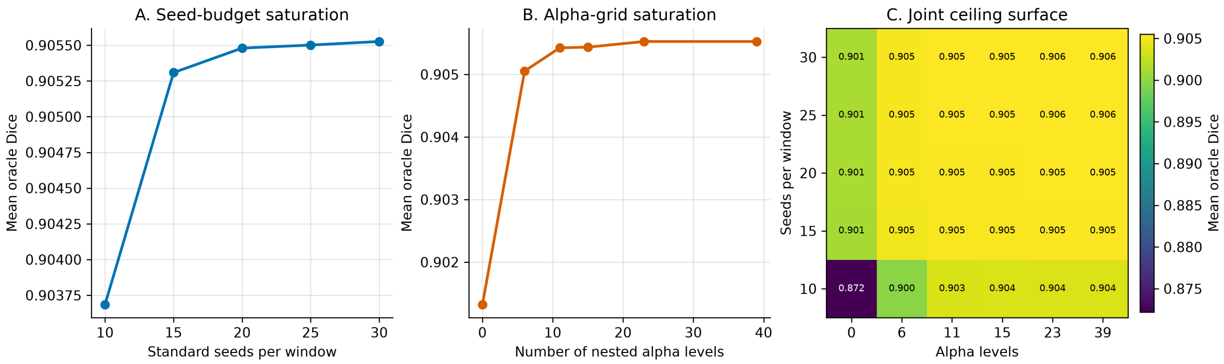

Figure 4: **Practical saturation of candidate capacity.** Seed and fuzzy α-cut budgets were enlarged in nested fashion. Every value is a retrospective pool-oracle mean Dice and is unavailable to the label-free selector.

contrast, the partial associations remained negative (−0.325, −0.321, and −0.319, respectively), with confidence intervals excluding zero. Pairwise confidence intervals for cross-cohort differences in the unadjusted correlations included zero.

Table 7: Association between weak-boundary fraction and pool-oracle Dice.

Cohort	N	Spearman ρ	Bootstrap 95% CI
TCGA-LGG	110	−0.537	[−0.668, −0.379]
UCSF-PDGM grade 2–3	99	−0.438	[−0.586, −0.259]
BraTS 2023 GLI	150	−0.467	[−0.593, −0.321]

The replicated direction supports the interpretation that weak, infiltrative FLAIR boundaries constrain the ability of this candidate family to represent an accurate contour. The association persisted beyond measured size, shape and local contrast differences, but remains observational and does not demonstrate that weak gradients causally determine segmentation failure.

4.5 External Validation and Cross-Grade Stress Test

Without parameter adjustment, the primary method achieved a mean Dice of 0.7124 (95% CI $[0.6491, 0.7724]$), median Dice of 0.8662, and 11 zero-Dice cases in the 99-case UCSF-PDGM grade 2–3 cohort (Table 8). Mean Dice was 0.6433 for grade 2 and 0.8025 for grade 3. The exploratory grade 3–grade 2 difference was +0.159 (95% CI $[0.044, 0.272]$; Mann–Whitney $U = 1542$, $p = 0.017$; Cliff's $\delta = 0.281$). Because subgroup comparison was not a prespecified confirmatory endpoint, it is interpreted as evidence of greater difficulty in the lower-contrast grade 2 subgroup rather than as a definitive grade effect.

In the BraTS 2023 cross-grade stress test, the same fixed method achieved mean Dice 0.8227, median Dice 0.8927, and one zero-Dice case among 150 studies. The higher BraTS value should not be read as a cross-dataset ranking: BraTS is predominantly high-grade, skull-stripped and template-aligned, whereas TCGA and UCSF serve different clinical and preprocessing regimes. Its role is to show that the fixed method remained operable under a substantial domain and tumour-grade shift.

Table 8: Primary performance by prespecified cohort role.

Cohort and role	N	Mean Dice	Median Dice	Zero Dice
TCGA-LGG, development	110	0.7893	0.9134	6
UCSF-PDGM grade 2–3, external validation	99	0.7124	0.8662	11
BraTS 2023 GLI, cross-grade stress test	150	0.8227	0.8927	1

4.6 Selector Robustness and Conditional ROI Analysis

The persistence exponent showed local tolerance but not unrestricted robustness. Changing γ from 1 to 0.5 yielded mean Dice 0.7909, a paired difference of +0.0017 (95% CI $[-0.0180, 0.0230]$). Increasing it to 2 reduced mean Dice to 0.7477 (difference −0.0416, 95% CI $[-0.0796, -0.0084]$). Reducing the seed budget from ten to five per window lowered Dice to 0.7087 (difference −0.0806, 95% CI $[-0.1277, -0.0396]$), while retaining only the narrow window lowered it to 0.7634 (difference −0.0259, 95% CI $[-0.0545, -0.0039]$). These analyses indicate that performance is relatively stable near the chosen persistence exponent, whereas candidate diversity and the two-window design are consequential.

The conditional brain-ROI renormalization was activated in five skull-intact TCGA cases. As a secondary development analysis, it increased mean Dice from 0.7893 to 0.8180 and reduced zero-Dice cases from six to two. The paired mean difference was +0.0288 (95% CI $[0.0013, 0.0631]$); four cases improved, one deteriorated and 105 were unchanged. Because both the correction and its adoption condition were developed on TCGA, this value is not the primary result and is not independent validation. The condition was never activated in UCSF-PDGM or BraTS 2023, both of which are skull-stripped.

In the threshold audit, changing the ROI-candidate requirement from 0.40 to 0.60 did not alter adoption because every eligible M_1 had $f_{\text{in}}(M_1) = 1$. Raising the first-mask threshold from 0.55–0.60 to 0.65, 0.70, and 0.75 increased adoptions from 4 to 5, 7, and 9, while deteriorations increased from 0 to 1, 3, and 4. The frozen 0.65 value is thus a developmental operating point rather than a theoretically unique constant. No upper-quartile large tumour triggered the frozen gate; one upper-quartile peripheral tumour triggered it and improved. These counts are too small to establish anatomical safety, and the single deterioration among the five frozen adoptions was substantial (Dice change −0.312).

4.7 Matched Training-Free Comparisons

Under the identical TCGA cases, selected slices, FLAIR input and whole-tumour target, Otsu thresholding achieved mean Dice 0.2429, whereas the AUCseg-BT reimplementation achieved 0.3315 (Table 9). The primary method exceeded Otsu by 0.5464 Dice (95% CI $[0.4936, 0.5964]$) and was better in 103 cases. Its paired difference from the AUCseg-BT reimplementation was 0.4578 (95% CI $[0.3844, 0.5297]$); it was better in 89 cases and tied in five.

Table 9: Matched training-free comparisons on TCGA-LGG. AUCseg-BT denotes the whole-tumour module reimplemented for the present single-FLAIR 2D protocol, not the complete original multiparametric 3D system.

Method	Mean Dice	Median Dice	Zero Dice	Dice ≥ 0.70
Otsu	0.2429	0.2303	3	0
AUCseg-BT reimplementation	0.3315	0.1551	40	34
Primary method	0.7893	0.9134	6	88

These are matched-protocol controls, not literature-wide superiority claims. They isolate the value of diversified score-topology candidates and label-free selection relative to simpler training-free intensity procedures under the same experimental unit. Comparisons with methods using additional modalities, volumetric context, supervision or prompts remain descriptive rather than rank-based.

4.8 Principal Interpretation

Taken together, the results support error decomposition rather than a single accuracy narrative. The primary method localized the lesion in 106 cases and represented a Dice-$\geq$ 0.70 boundary in 103, yet failed to recover an available candidate by at least 0.10 Dice in 16 cases. Enlarging the candidate family increased its practically saturated ceiling to approximately 0.906 but did not increase label-free delivery. Candidate generation is therefore not exhausted in the few localization- and boundary-limited cases, while selection is the more prominent remediable bottleneck among representable cases. A future selector trained with labels could potentially exploit this capacity, but it would define a supervised or hybrid method and should not be described as training-free.

5 Limitations

Several limitations define the scope in which these findings should be interpreted. First, TCGA-LGG served simultaneously as the development and internal evaluation cohort. Reference masks were used to select the largest-tumour slice, determine fixed parameters, quantify performance, assign error classes, and calculate candidate-pool ceilings. Although candidate generation and final selection were label-free for each fixed slice, the reported TCGA performance is not an unbiased estimate from a held-out test cohort. The UCSF-PDGM grade 2–3 analysis provides independent institutional validation of the fixed primary method, but it does not remove the risk of development-set optimism in individual design choices. Parameter comparisons, including the persistence exponent, window configuration, seed budget and ROI adoption condition, should therefore be regarded as developmental analyses rather than confirmatory evidence of optimality. The ROI threshold audit further showed that the number of deteriorations increased as the first-mask adoption condition was relaxed, and one of the five frozen-gate adoptions lost 0.312 Dice. The absence of a deterioration in the very small retrospective large/peripheral strata is insufficient to establish anatomical safety.

Second, the experimental unit was a single axial slice selected using the reference tumour mask. The study consequently evaluates segmentation conditional on a known tumour-bearing slice and does not address automatic lesion detection, slice selection, multifocal sampling across a volume, or three-dimensional boundary consistency. Selecting the largest-tumour slice also favours sections with greater lesion representation and may underestimate the difficulty of small or weakly expressed tumour sections. Extension to volumetric FLAIR should be evaluated with patient-level separation, automatic study-level localization and three-dimensional surface measures before clinical or workflow-level claims are made.

Third, the method uses only FLAIR intensity and image-derived geometry. This restricted input is central to the intended low-resource setting, but it excludes complementary information from T1-weighted, contrast-enhanced T1-weighted and T2-weighted MRI. Weak or infiltrative boundaries may therefore remain intrinsically ambiguous in the available observation. The practically saturated oracle Dice of approximately 0.906 is an upper bound only for the finite candidate family examined here; it is neither automatic segmentation performance nor a universal single-FLAIR ceiling. A different handcrafted, learned or multimodal generator could exceed it. Conversely, the gap between this ceiling and the delivered mask cannot be assumed recoverable without labels: the present quality criterion did not convert the additional fuzzy α-cut capacity into higher label-free performance.

Fourth, external evaluation spans only one independent lower-grade cohort. UCSF-PDGM and BraTS 2023 are skull-stripped, whereas the TCGA derivative is skull-intact and differently preprocessed. The BraTS result is therefore a cross-grade and preprocessing-domain stress test, not lower-grade external validation, and cohort means should not be interpreted as direct rankings of tumour grades or datasets. The conditional brain-ROI correction was developed on TCGA and was inactive in both external cohorts; its apparent reduction of complete failures requires confirmation in an independent skull-intact lower-grade cohort. Differences in annotation conventions, acquisition protocols and spatial resolution may also affect overlap and boundary measures. Surface distances reported in pixels are not interchangeable with physical millimetres unless image spacing is harmonized.

Finally, the error decomposition, boundary analyses and qualitative case panels are retrospective. The successful examples were selected using reference-tumour area strata and retrospective Dice proximity to the class median; they illustrate typical within-class behaviour rather than prospective case selection or an independent performance estimate. The thresholds defining localization, boundary representation and selection failure are operational research definitions rather than clinical acceptance criteria. Although the 27-setting sensitivity grid preserved the qualitative dominance of selection failures, the exact class counts changed with these definitions, especially when the oracle threshold was raised to 0.80. The negative weak-boundary association replicated in direction across all three cohorts, and none of the pairwise confidence intervals for differences in Spearman correlation excluded zero; this supports directional consistency but does not establish equal effect magnitudes. Because the analysis is observational and derives boundary descriptors from the reference contour, it does not prove that weak gradients causally produce low candidate ceilings. Larger prospective multi-institutional cohorts, prespecified primary endpoints and externally fixed decision rules are required to test causal or clinical claims.

6 Conclusions

This study demonstrates that performance in training-free single-FLAIR lower-grade glioma segmentation is more informative when decomposed into localization, boundary representation and label-free selection. On the TCGA-LGG development cohort, the primary method achieved a mean Dice of 0.789, localized the tumour in 106/110 cases and contained a Dice-$\geq$ 0.70 candidate in 103/110. Among these 103 representable cases, however, 16 retained an oracle–selected difference of at least 0.10. The resulting distinction is fundamental: failure to generate an adequate boundary and failure to select an already available boundary are separate error mechanisms and require different remedies.

The candidate analyses reinforced this conclusion. The primary seeded superlevel-set family had an oracle Dice of 0.872, while nested seed and fuzzy α-cut expansion reached a practically saturated ceiling of 0.906. Despite this increase in candidate capacity, label-free delivered performance did not improve. More extensive candidate generation is therefore useful for the remaining localization- and representation-limited cases, but it does not resolve the ranking problem in representable cases. Consistently, a larger weak-boundary fraction was associated with a lower attainable ceiling in TCGA-LGG, UCSF-PDGM and BraTS 2023, supporting boundary salience as a reproducible correlate of representability without implying causation.

With all parameters fixed after TCGA development, the method achieved mean Dice 0.712 on the independent UCSF-PDGM grade 2–3 cohort and 0.823 in the BraTS 2023 cross-grade stress test. These results support transfer of the fixed-slice method across institutions and preprocessing regimes, but not direct cross-cohort ranking or volumetric clinical performance. The secondary brain-ROI correction reduced TCGA complete failures, yet remains to be validated in an independent skull-intact lower-grade cohort. Likewise, the present evidence does not isolate an accuracy advantage of NewMetric over alternative edge-preserving operators; its role here is to provide the fixed geometric field within which candidate generation and selection were studied.

The principal contribution is therefore both methodological and diagnostic: a training-free segmentation result should report not only its delivered overlap, but also whether the lesion was localized, whether an adequate boundary existed within the candidate family, and whether the label-free criterion recovered it. For the present single-FLAIR setting, the most immediate opportunity lies in improving candidate selection while preserving the distinction between training-free and label-trained decision rules. Future work should evaluate that decision problem under prespecified patient-level splits, extend the method to automatic three-dimensional analysis, and validate all fixed rules in additional independent lower-grade cohorts.

Author Contributions

S.C.: conceptualization, theoretical formulation, methodology, formal analysis, investigation, validation, and supervision. H.K.: software, implementation, visualization, and writing of the original manuscript. Both authors reviewed and approved the final manuscript.

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Conflict of Interest

The authors declare no competing interests.

Ethics Statement

This study used only de-identified retrospective imaging datasets obtained through their applicable distribution channels and involved no new participant recruitment, intervention, or collection of identifiable information. Ethical approval and informed-consent procedures for the source cohorts were governed by the institutions that originally collected and released those datasets. The present analysis introduced no new contact with participants and used the datasets under their applicable access and data-use conditions.

Data Availability

TCGA-LGG, BraTS 2023 GLI, and UCSF-PDGM are available through their respective controlled or public distribution channels under the terms specified by the data providers. The present study does not redistribute the source MR images or reference masks. Derived case-level measurements and the fixed experimental specification are archived with the accompanying research materials.

Code Availability

The fixed method, data-preparation procedures, evaluation routines, hyperparameters, and manuscript sources are archived in a private GitHub repository at `https://github.com/salim-ceyhan/flair-lgg-error-decomposition/tree/manuscript-release`. Repository access can be provided confidentially to editors and reviewers on request. The repository will be made publicly accessible upon acceptance of the article; source imaging data are not included.

A Geometric Role of the Edge-Preserving Field

The geometric component of the method is confined to construction of the edge-preserving field F_n used by the score in Eq. (5). Following the spatial-feature-manifold formulation [32], a scalar image may be represented as the graph

$$X(x^1, x^2) = \big(x^1, x^2, I(x^1, x^2)\big), \tag{11}$$

whose induced metric couples spatial displacement to intensity variation. The NewMetric operator used here introduces a direction-dependent deformation of this metric and performs a short diffusion with fixed parameters $(\beta_{\mathrm{NM}}, \Delta t, n) = (5.0, 0.15, 3)$ [5]. The numerical result is then normalized to produce F_n; no Gaussian filtering is applied in the primary method.

This construction should be interpreted narrowly. NewMetric supplies the fixed field from which edge permeability, intensity weighting and candidate boundaries are derived. The study does not identify NewMetric itself as the source of the observed Dice performance, nor does it establish superiority over other edge-preserving diffusion operators. The empirically supported contribution is the decomposition of error within the complete training-free method. Accordingly, “Finsler” describes the geometric origin of the image field rather than a stand-alone accuracy claim.

B Seeded Persistence and Candidate Stability

For a score field $S : \Omega \to [0, 1]$, define its superlevel sets $X_t = \{x \in \Omega : S(x) \geq t\}$. As the threshold decreases, these sets form a nested filtration. Given a seed s, the method follows the connected component $C_t(s) \subseteq X_t$ containing that seed. A candidate is recorded when the component area remains within the relative tolerance specified in Section 3.4; the corresponding threshold duration is its persistence $\pi(M)$.

This quantity is a seed-conditioned, zero-dimensional stability descriptor. It measures how long a connected region remains approximately unchanged before growth or merging, but it does not identify that region as tumour. A strictly increasing transformation of S preserves the ordering and connectivity of the superlevel sets, although a nonlinear transformation need not preserve the raw numerical width of a threshold interval. Persistence is therefore used as a within-image relative weight rather than as an absolute quantity comparable across images or cohorts. Its empirical contribution is evaluated rather than assumed: nearby exponents performed similarly, whereas excessive persistence weighting degraded accuracy.

C Supplementary Verified Analyses

C.1 Selector and Candidate-Diversity Sensitivity

Table 10 summarizes the prespecified perturbations of the frozen TCGA candidate records. The result was tolerant to a moderate change in the persistence exponent, but not to excessive weighting, removal of half of the seed budget, or removal of the broad intensity window. These comparisons are development-set sensitivity analyses and are not independent tests of parameter optimality.

Table 10: Sensitivity of label-free selection on TCGA-LGG. Differences are paired against the primary configuration.

Configuration	Mean Dice	Difference	Bootstrap 95% CI
Primary: $\gamma = 1$, ten seeds, two windows	0.7893	—	—
$\gamma = 0.5$	0.7909	+0.0017	[−0.0180, 0.0230]
$\gamma = 2$	0.7477	−0.0416	[−0.0796, −0.0084]
Five seeds per window	0.7087	−0.0806	[−0.1277, −0.0396]
Narrow window only	0.7634	−0.0259	[−0.0545, −0.0039]

C.2 Conditional ROI Renormalization

The ROI correction altered only the skull-intact TCGA cohort (Table 11). It was adopted in five cases, improving four and worsening one; all other cases were unchanged. Its absence of activation on skull-stripped data demonstrates consistency with its intended trigger, not external validation of the correction itself.

Table 11: Secondary analysis of conditional brain-ROI renormalization.

Cohort	Base Dice	ROI-conditional Dice	Zero Dice	Adopted
TCGA-LGG	0.7893	0.8180	$6 \to 2$	5/110
UCSF-PDGM grade 2–3	0.7124	0.7124	$11 \to 11$	0/99
BraTS 2023 GLI	0.8227	0.8227	$1 \to 1$	0/150

C.3 Interpretation Boundaries

All appendix analyses preserve the distinctions used in the main text. Oracle values require reference masks and quantify only the capacity of a specified candidate family. TCGA comparisons are developmental. UCSF-PDGM is the sole independent lower-grade external-validation cohort, whereas BraTS 2023 is a cross-grade stress test. No appendix result changes the primary label-free Dice of 0.7893 or converts the two-dimensional, reference-selected-slice protocol into a fully automatic volumetric setting.

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